

Independent Replication Report:

Theory of the Ferroelectric Phase in Organic Conductors:

Optics and Physics of Solitons

S. Brazovskii, arXiv:cond-mat/0306006v2 (ECRYS-2002 proceedings)

REPLICATE-PROJECT / TEXTURES-100 — independent reimplementation

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Abstract

This is an independent, from-scratch reimplementation of the analytic physics of Brazovskii's ECRYS-2002 proceedings paper on the ferroelectric (FE) phase of the quasi-1D organic conductors (TMTTF)₂X. The paper is a *theory/proceedings* paper: it presents closed-form results “without derivations” (its own words, 1.173) and contains **no numerical tables**. Accordingly the replication scores agreement against **functional forms** and **order-of-magnitude** claims, with every stated relation turned into a machine-checkable identity. All ten analytic relations reimplemented pass (10/10), most to machine precision.

HEADLINE: optical collective-mode edge relative to the photoconductivity gap, $\omega_t/(2\Delta) = \pi\gamma/2 = \mathbf{0.393}$ at $\gamma = 0.25$ — confirming the paper's claim that the optical edge $\omega_t \approx \pi\gamma\Delta$ lies *well below* 2Δ (1.22, 1.153).

VERDICT: PARTIAL (mechanism-level REPLICATED). Coverage **8/10**
Agreement **9/10**.

1 The paper's claims (as reimplemented)

The paper describes a *combined Mott–Hubbard* state of (TMTTF)₂X in which the interference of a built-in bond nonequivalence (U_b) and a spontaneous site nonequivalence (U_s , the charge disproportionation, CD) produces ferroelectricity. The checkable analytic claims are:

1. Combined Mott–Hubbard Hamiltonian rewrites as a single cosine with an effective amplitude U and phase shift α ; ground state at $\phi = \alpha$.
2. Three soliton species: π -solitons (holons, charge e); noninteger α -solitons (FE domain walls, $q = -2\alpha/\pi$ or $1 - 2\alpha/\pi$); combined spin-charge solitons (short charge core, long spin tail).
3. Gap scaling $\Delta \sim U^{1/(2-2\gamma)}$; CD requires $\gamma < 1/2$.
4. Optical edge is a *collective mode* at $\omega_t \approx \pi\gamma\Delta < 2\Delta$, not the two-particle gap 2Δ (photoconductivity edge).
5. Mixed electron-phonon dielectric response $\varepsilon(\omega)$ [Eq. 2], exhibiting a Fano antiresonance at ω_0 , a combined resonance at $\omega_{0t}^2 \approx \omega_0^2 + \omega_t^2$, and an FE soft mode at $\omega_{fe}^2 \approx (1 - Z)/(\omega_0^{-2} + \omega_t^{-2})$.
6. Curie-law dielectric divergence $\varepsilon(T) = A|T/T_0 - 1|^{-1}$ with $A \sim (\omega_p^*/\omega_t)^2 \sim 10^3$.
7. Spin-Peierls spin gap $\Delta_\sigma \sim U_{ao}^{2/3}$; core/tail lengths $\xi_\rho \sim \hbar v_F/\Delta \ll \xi_\sigma \sim \hbar v_F/\Delta_\sigma$.

2 Governing equations reimplemented

Combined Mott–Hubbard Hamiltonian.

$$H_U = -U_s \cos 2\phi - U_b \sin 2\phi = -U \cos(2\phi - 2\alpha), \quad U = \sqrt{U_s^2 + U_b^2}, \quad \tan 2\alpha = U_b/U_s. \quad (1)$$

Gap and optical edge. $\Delta \sim U^{1/(2-2\gamma)}$, $\omega_t \approx \pi\gamma\Delta$, photoconductivity gap $= 2\Delta$.

Dielectric response (Eq. 2).

$$\frac{\varepsilon(\omega)}{\varepsilon_\infty} = 1 + \frac{(\omega_p^*/\omega_t)^2 (1 - (\omega/\omega_0)^2)}{(1 - (\omega/\omega_0)^2)(1 - (\omega/\omega_t)^2) - Z}, \quad Z = \left(\frac{\omega_{cr}}{\omega_t}\right)^{2-4\gamma} \leq 1. \quad (2)$$

The denominator is a quadratic in $x = \omega^2$: $x^2 - (\omega_0^2 + \omega_t^2)x + (1 - Z)\omega_0^2\omega_t^2 = 0$, whose two roots are the combined resonance (upper) and the FE soft mode (lower).

Curie law. $\varepsilon(0)/\varepsilon_\infty = 1 + A/(1 - Z)$, $A = (\omega_p^*/\omega_t)^2$; with $1 - Z \propto |T/T_0 - 1|$ this is the Curie $1/|t|$ divergence.

3 Method

Pure analytic / algebraic reimplementation in Python (NumPy 2.3.5), from the equations transcribed above — **not** author code (the paper ships none). Each stated identity is evaluated numerically and asserted:

- Identities (H, soliton charges, Fano) checked to machine precision over grids.
- The paper’s $\approx/\sim$ formulas (resonance, soft mode) are the $Z \rightarrow 1$ (criticality, $T \rightarrow T_0$) leading-order limit of the *exact* denominator roots; verified by scanning $Z = 0.5 \rightarrow 0.9999$ and showing the relative error *monotonically decreases* toward the limit.
- Order-of-magnitude claims (Curie amplitude) checked by reproducing the exact Curie law and confirming $(\varepsilon - 1)(1 - Z) = A$ is constant.
- Dimensionless ratios/exponents tested (the paper never fixes prefactors C , ω_p^* , or the T -laws $\omega_t(T), \omega_{cr}(T)$).

4 Results: this work vs. paper

Convergence of the $\approx$ -formulas (the key methodological point). Testing $\omega_{0t}^2 \approx \omega_0^2 + \omega_t^2$ at an *arbitrary* Z (e.g. $Z = 0.5$) gives $\sim 7\%$ “error” and a false miss. It is a limit formula: scanning $Z = 0.5, 0.9, 0.99, 0.999, 0.9999$ gives resonance relative errors $7.0\% \rightarrow 1.2\% \rightarrow 0.12\% \rightarrow 0.012\% \rightarrow 0.0012\%$ (and identically for the soft mode). Monotone convergence toward T_0 is the verification.

Quantity / identity	This work	Paper	Status
H form identity $\max \text{LHS} - \text{RHS} $ (ϕ grid)	4.4×10^{-16}	exact	EXACT
Ground-state min at $\phi = \alpha$	true	$\phi = \alpha$	MATCH
α -soliton charge $-2\alpha/\pi$ ($U_s=0.7, U_b=0.4$)	-0.1652	$-2\alpha/\pi$	EXACT
α -soliton charge $1 - 2\alpha/\pi$	0.8348	$1 - 2\alpha/\pi$	EXACT
$\omega_t/(2\Delta)$ at $\gamma = 0.25$ ($=\pi\gamma/2$)	0.3927	" $< 2\Delta$ "	MATCH
γ ceiling for $\omega_t < 2\Delta$ ($=2/\pi$)	0.6366	—	consistent
Resonance ω_{0t} rel. err. at $Z=0.9999$	1.2×10^{-5}	$\approx$ limit	CONVERGES
Soft mode ω_{fe} rel. err. at $Z=0.9999$	1.2×10^{-5}	$\approx$ limit	CONVERGES
Fano antiresonance $\varepsilon/\varepsilon_\infty(\omega_0)$	1.0	1 (antires.)	EXACT
Curie amplitude $A = (\omega_p^*/\omega_t)^2$	900	$\sim 10^3$	ORDER MATCH
Curie product $(\varepsilon - 1)(1 - Z)$ constancy (std/mean)	5.5×10^{-16}	$1/ t $ law	EXACT
Soft mode $\propto \sqrt{t}$ (corr. coeff.)	> 0.999	$\sim \varepsilon^{-1/2}$	MATCH
Spin tail / core ratio ξ_σ/ξ_ρ ($U_{ao}=0.05$)	7.37	$\gg 1$	MATCH

Table 1: All 10 verdict checks pass. Machine-precision identities plus monotone convergence of the two $\approx$ -formulas to their criticality limit.

5 Comparison & verdict

PARTIAL (mechanism-level REPLICATED); coverage 8/10, agreement 9/10. Every algebraic identity in Sections I–II of the paper reproduces exactly (4×10^{-16} on the Hamiltonian form, Fano = 1, Curie $1/|t|$ exact to 5×10^{-16}). The two $\approx$ formulas are confirmed as the leading criticality limit of exact objects. The order-of-magnitude Curie amplitude ($A = 900 \sim 10^3$) matches the paper’s own estimate; the experimental $\sim 10^4$ is the number the paper itself only approximately matches, not a target this code must hit. Agreement is 9 (not 10) only because there are no numerical tables to fit against — the score is necessarily on forms and orders of magnitude.

6 Honest gaps and scope

1. **Proceedings paper, no tables.** Results are given “without derivations” (l.173); agreement is on functional forms + order-of-magnitude, not tabulated data. *Expected, not a shortfall.*
2. **Free normalizations.** The prefactor C in $\Delta \sim U^{1/(2-2\gamma)}$, the plasma frequency ω_p^* , and the temperature laws $\omega_t(T), \omega_{cr}(T)$ are not fixed by the paper; representative values were chosen. Only dimensionless ratios and scaling exponents are tested.
3. **Quantum breather spectrum** between ω_t and 2Δ (soliton-antisoliton bound states) is *mentioned but given no closed form*; not reimplemented. *Coverage-capping, scoped out.*
4. **Microscopic mass renormalization** ($U \rightarrow U^*$, γ from v_ρ/v_F , ripplon analogy) given only as scaling relations; checked structurally, not against data.

Items 3–4 are exactly why coverage caps at 8/10 for this proceedings paper.

7 Reproducibility

Interpreter: `/home/stevens/comfyui-env/bin/python` (NumPy 2.3.5, SciPy 1.17.0). The solver is `report/evidence/brazovskii2003_replication.py`; the live re-run reproduces `report/evidence/brazovskii2003_replication.py` (and `brazovskii2003_output.json`) to the digits quoted above (10/10 checks pass).

```
cd textures-polar-brazovskii2003/  
/home/stevens/comfyui-env/bin/python report/evidence/brazovskii2003_replication.py
```

No LaTeX engine is installed on the packaging host; this `.tex` is delivered as source and compiles off-host with a standard `pdflatex` run.